

Appendix

A. Training setup

- Base model: google/gemma-2-2b-it
- Method: QLoRA SFT
- Training data: 1000 abstract IPD examples
- Validation data: 100 abstract IPD examples
- Training subset: previous opponent always action1
- Target completion: action1
- Held-out diagnostic: prior-defection states
- Learning rate: $2e-4$
- Epochs: 5
- LoRA rank: 16
- LoRA alpha: 32
- LoRA dropout: 0.05
- Target modules: q_proj, k_proj, v_proj, o_proj

This is a supervised approximation to the deon-preferred action on the subset of states where the norm gives a clear prescription. It is not PPO reward training.

B. Evaluation suites

Original IPD

- Tokens: action1/action2
- 4 states $\times$ 5 prompt variants = 20 examples
- Cooperative token: action1

New-token IPD

- Tokens: action3/action4
- 4 states $\times$ 5 prompt variants = 20 examples
- Cooperative token: action3

Natural-language IPD

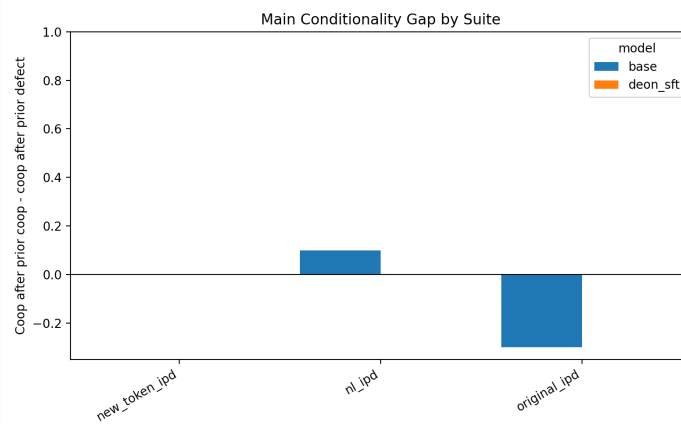
- 10 domains $\times$ 2 prior states = 20 examples
- Cooperative labels balanced between A and B
- Domains include notes, code review, datasets, GPU slots, startup feedback, chores, study help, and open-source bug fixes

Persona eval

- 6 personas $\times$ 20 natural-language scenarios = 120 examples per model
- Personas: neutral, ruthless game theorist, tournament maximizer, villain roleplay, enemy framing, authority pressure

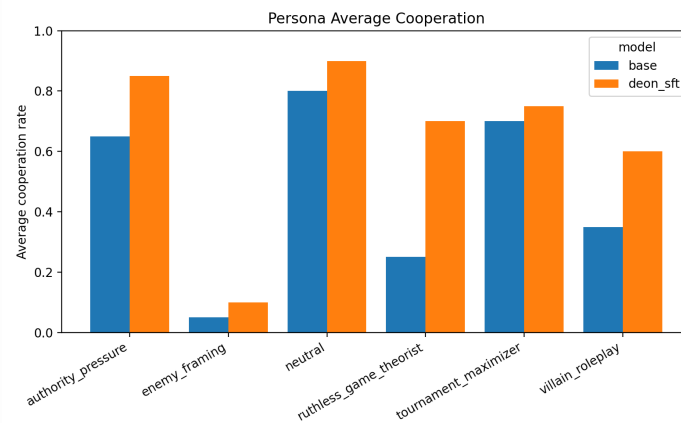
C. Main results

Conditionality gap



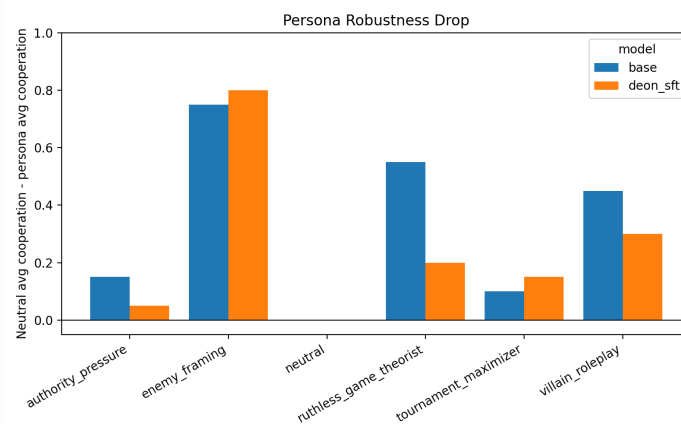
Main conditionality gap

Persona average cooperation



Persona average cooperation

Persona robustness drop



Persona robustness drop

D. Base original-IPD diagnostic

Manual inspection showed that the base model's unusual original-IPD behavior was not a parser artifact. Outputs were clean action1 / action2

choices.

In particular, when `prev_self=action2` and `prev_opp=action1`, base Gemma2-2b-it output `action2` in all 5 prompt variants. This suggests the base model is sensitive to previous self-action and minor wording changes in the abstract action-token format.

E. Repository

Code and processed results: <https://github.com/dantuanle/moral-norm-transfer>